

Executive Proposal: Standardizing Media Metadata Extraction via In-House Engine

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1. Executive Summary

Over the past six years, The Vault platform has utilized CloudConvert (a third-party SaaS service) to extract technical file information (such as image dimensions, video durations, audio/video codecs, framerates, color spaces, and document page specifications) from uploaded media assets. This information is saved directly to our database and powers downstream processes such as asset validation, automated layout rendering, and media delivery.

Operating an external third-party service for metadata extraction over an extended period has introduced key operational challenges:

- **Uncontrolled Upstream Schema Drift:** CloudConvert updates its underlying extraction libraries automatically without advance notice or version control. These updates have introduced subtle changes to metadata property names over time, requiring our engineering team to continually write complex backwards-compatibility code (such as checking multiple variations of dimensions like ImageWidth, width, ImageSize, and MaxPageSizeW) to ensure older campaign assets do not break modern rendering workflows.
- **Operational Latency & Webhook Dependency:** External API roundtrips and remote queuing routinely add between 2 and 28 seconds of latency per asset, keeping Lambda functions alive longer and introducing an external network dependency into core ingestion.
- **Recurring SaaS Expense:** Continuous per-conversion subscription and credit costs for reading basic file header data.

Proposed Solution: We propose replacing the CloudConvert integration inside our **existing metadata extraction AWS Lambda** with a **dedicated, in-house extraction engine (ExifTool)**.

Because the Lambda infrastructure is already provisioned and executes asynchronously, infrastructure compute costs remain virtually unchanged, while completely eliminating external SaaS billing. To further improve resilience, we recommend introducing a standard message queue (Amazon SQS) between the backend and Lambda to buffer high-volume upload bursts.

2. Ingestion Architecture Workflow

1. [User / Frontend] —(Direct File Upload)—> [Amazon S3] 2. [Frontend] —(Upload Complete API Call)—> [Backend Service] 3. [Backend Service] —(Enqueue Task)—> [Amazon SQS Queue Buffer] 4. [Amazon SQS Queue] —(Asynchronous Trigger)—> [Existing AWS Lambda (In-House Engine: <400ms)] 5. [Existing AWS Lambda] —(Direct Database Save)—> [PostgreSQL files.metadata]

3. Forensic Analysis: 6-Year Schema Drift & Database Impact

A forensic database audit of **593,504 files** in the database revealed how CloudConvert's silent upstream tool upgrades generated distinct generations of conflicting metadata structures:

GENERATION	ENGINE VERSION RECORDED	ASSET COUNT	DATE RANGE	IMPACT ON DATABASE RECORDS
Generation 1	v12.36	18,029	Feb 2022 – Feb 2024	Early CloudConvert worker version (decimal duration formats, early bitrate tags)
Generation 2	v12.56	271,126	Nov 2021 – Aug 2026	CloudConvert container upgrade (timecode duration shifts, enhanced vector tags)
Generation 3	NO_EXIFT00L_TAG	272,602	Mar 2022 – Aug 2026	S3 header-only or non-standard extractions

Real-World Discrepancy Case Studies from Historical Data

Case 1: Video Duration Format Shift (Decimal Seconds vs. Timecodes)

In earlier uploads, video duration was stored as a decimal string with unit ("20.00 s"). Following upstream tool updates, the format shifted to a timecode string ("0:00:30"). Standard numerical parsing functions (parseFloat) evaluate "0:00:30" as 0, causing downstream rendering engines to miscalculate asset length unless complex string-splitting fallback logic was applied.

- File #259754 (S0_Enterprise_C12_Rent_A_Car_ENG_Molde_1.mp4, 2024-09-19): Duration: "20.00 s"

- File #475028 (Allstate_C1_Generic_ATX_Base_South.mp4, 2026-01-27):
Duration: "0:00:30"

Case 2: Bitrate Tag Splitting (AvgBitrate vs. Bitrate vs. VideoBitrate)

Older systems looked for a single Bitrate field. Newer MP4/MOV extractions exclusively output AvgBitrate, while MPEG videos output VideoBitrate. Across 142,287 MP4 videos, 117,445 files contain AvgBitrate, while only 8 files contain Bitrate. Code querying metadata.Bitrate returned empty values for over 99% of video assets.

Case 3: Adobe Illustrator Canvas Dimensions (MaxPageSizeW vs. ImageWidth)

Vector artwork files (.ai / .eps) do not contain a pixel grid. ExifTool returns ImageWidth: null and instead stores the true artboard canvas dimensions inside MaxPageSizeW and BoundingBox. Across 2,099 Adobe Illustrator master assets, ImageWidth is completely null.

- File #381646 (S0_P&G_Gillette_C2_SharedStatic_Base_Artwork_1.ai):
ImageWidth: null, MaxPageSizeW: "242", BoundingBox: "-4 -263 1746 14"
- File #307768 (S0_UEFA_C6_VAR_Base_144px_(246m)_1.ai): ImageWidth: null,
MaxPageSizeW: "18800"

Case 4: Document Page Count Tag Changes (PageCount vs. Pages)

Early PostScript files recorded document lengths under Pages, whereas standard PDF extractions output PageCount.

- File #308888 (Brax_C15_SUPERBET_ISG_Virtual_Carpets_All_Venues_1.pdf):
PageCount: "1", Pages: null
- 30 legacy PostScript documents contain Pages: 1

Case 5: PNG Resolution & DPI Inconsistencies (XResolution vs. PixelsPerUnitX)

Image resolution is recorded either under XResolution (in inches) or raw pixel density PixelsPerUnitX (in meters) depending on the authoring software and upstream parser state.

- File #562777 (1XBET_C11_222_BA2_BGR.png): XResolution: "144",
ResolutionUnit: "inches", PixelsPerUnitX: "5669"

- File #546446 (AFC_Commercial_Team_C4_Ca3_FT_5.png): XResolution: null, PixelsPerUnitX: "2835"

Case 6: Photoshop Layer Data & Color Spaces (PSD Assets)

Master Photoshop PSD files require layer count and color space data for automated rendering. Upstream changes varied between returning raw color space integers vs. descriptive strings.

- File #393594 (National_Partners-Shared_Artwork_C9_V2_Miele_Base_144_1.psd): LayerCount: "4", ColorSpace: "sRGB"
- File #377806 (S0_Apple_C7_Stick_Group_2_Base_2.psd): LayerCount: "11", ColorSpace: "Uncalibrated"

4. Proof of Concept (POC) Validation & Benchmark Results

A Proof of Concept was developed to verify data parity between live CloudConvert API responses and the in-house extraction engine across representative media formats in The Vault catalog:

MEDIA FORMAT	ASSET TYPE	EXTRACTED TECHNICAL ATTRIBUTES	VERIFICATION STATUS
PNG	Brand graphics, transparent overlays	Dimensions, Bit Depth, Color Type, DPI, Gamma	100% Parity
JPEG	Photography, campaign images	Dimensions, EXIF metadata, Megapixels, Color Space	100% Parity
MP4	Video commercials, board renders	Resolution, Duration, Framerate, Bitrate, Codecs	100% Parity
QuickTime MOV	High-definition master videos	Resolution, TimeScale, Duration, Codec profile	100% Parity
PDF	Print proofs, campaign specifications	Page Count, PDF Version, Creator Software	100% Parity
Adobe Illustrator (AI / EPS)	Vector artwork, signage dimensions	Artboard Dimensions, Bounding Box, XMP metadata	100% Parity
Adobe Photoshop (PSD)	Multi-layer master graphics	Layer Count, Blend Modes, Layer Names, ICC	100% Parity
WebP	Web banners	Dimensions, Compression mode, Alpha channel	100% Parity

Live Real-Time Benchmark Comparison

BENCHMARK DIMENSION	CLOUDCONVERT SAAS (LIVE API)	IN-HOUSE ENGINE (PROPOSED)	OPERATIONAL ADVANTAGE
Processing Latency	~28,990 ms (Upload + Remote Queue + Network)	381 ms (Direct stream parse)	76x Faster Extraction

BENCHMARK DIMENSION	CLOUDCONVERT SAAS (LIVE API)	IN-HOUSE ENGINE (PROPOSED)	OPERATIONAL ADVANTAGE
Critical Field Match Rate	100%	100%	Exact Match
Width & Height Accuracy	Identical	Identical	100% Match
File Format & MIME Type	Identical	Identical	100% Match
Color Profiling & Bit Depth	Identical	Identical	100% Match
Third-Party SaaS Cost	Recurring monthly credit billing	\$0.00	Complete Cost Elimination

5. Architectural Enhancements & Stability

- **Burst Upload Protection:** During large campaign launches, Amazon SQS buffers incoming jobs to ensure the Lambda processes files steadily without database connection spikes.
- **Automated Error Handling:** Corrupt or malformed files route to a Dead-Letter Queue (DLQ) for alerting without blocking remaining assets.
- **Enhanced Security:** Media assets remain entirely inside our private AWS environment.

6. Strategic Business Impact Summary

STRATEGIC AREA	CURRENT STATE (CLOUDCONVERT SAAS)	PROPOSED STATE (IN-HOUSE ENGINE)	BUSINESS BENEFIT
Third-Party SaaS Cost	Recurring monthly credit billing	\$0.00 / month	Direct operational savings.
Infrastructure Cost	Existing Lambda	Existing Lambda (No change)	Zero added compute spend.
Processing Speed	2 to 28 seconds per asset	Under 400 milliseconds	Snappier UI and faster asset availability.

STRATEGIC AREA	CURRENT STATE (CLOUDCONVERT SAAS)	PROPOSED STATE (IN-HOUSE ENGINE)	BUSINESS BENEFIT
Data Consistency	Vulnerable to unannounced upstream changes	Fixed, version-controlled schema	Eliminates code complexity and maintenance.
Queue Stability	Direct invocation model	Message queue buffer (Amazon SQS)	Higher reliability during peak upload volumes.
Data Privacy	Files sent to external third-party servers	Contained within private AWS cloud	Full data privacy and compliance.

7. Conclusion

Replacing the external CloudConvert integration with an in-house engine inside our existing Lambda resolves the long-standing schema drift challenge, accelerates asset ingestion from multi-second queues to sub-second execution, eliminates recurring SaaS expenses, and introduces message queue buffering for peak upload stability.